Supplement - Reference-Set Decoupled Co-Adaptive Training for Physics-Informed Neural Networks

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1 Scope and evidence

The supplement preserves the complete V10 paired-contrast families. It introduces no new method-ranking evidence. Original V10 data are frozen at commit `4161ed57f232ea7b796480b38201502b3cabbf67`.

2 Observation regimes

Case	Sensors	Noise	Noise level	Outlier fraction	Missing pattern
sparse clean	13	none	0.0	0.0	none
sparse noisy	13	gaussian	0.1	0.05	none
extreme	1	heteroscedastic	0.2	0.05	none
block missing	64	correlated	0.05	0.0	time_block

Counts above are checked against stored run summaries. The code defines noise scaling and the missing-time-block construction in `src/recoa_pinn/problems/base.py`. All V10 comparisons use MSE for observations; the historical Huber recipes remain separately identified.

3 Inference and notation

Ten paired seeds are used: 11, 22, 33, 44, 55, 66, 77, 88, 99, 111. The exact two-sided sign test ignores zero differences. Percentile intervals use 10,000 paired bootstrap resamples and seed 9092026. Holm correction is applied to 16 contrasts separately for each metric and campaign. Intervals are marginal, not simultaneous. A nonsignificant difference does not establish equivalence.

Notation: F = full exponential; Q = full quadratic; G = global only; L = local only; N = neither; V = vRBA. Interaction = F - G - L + N on the additive error scale. L2 is relative solution error; max is the maximum absolute error on the declared grid, not a certified continuous-domain bound. Negative F-G and V-M contrasts favor F and V, respectively.

4 Difficult-observation ablations: L2

Case	Contrast	Median difference	Marginal 95% CI	Sign p	Holm p
extreme	F-L	0.00177	[-0.00424, 0.00659]	0.75391	1.00000
extreme	F-G	-0.06077	[-0.08657, -0.02842]	0.02148	0.32227
extreme	F-Q	-0.00411	[-0.01567, 0.01379]	0.75391	1.00000
extreme	F-G-L+N	0.00082	[-0.00372, 0.00242]	1.00000	1.00000
sparse noisy	F-L	0.00354	[-0.00138, 0.00505]	0.10938	1.00000
sparse noisy	F-G	-0.08674	[-0.14774, -0.00986]	0.10938	1.00000
sparse noisy	F-Q	-0.01086	[-0.02904, 0.00601]	0.34375	1.00000
sparse noisy	F-G-L+N	0.00305	[-0.00081, 0.00723]	0.34375	1.00000
sparse clean	F-L	-0.00007	[-0.00625, 0.00318]	1.00000	1.00000
sparse clean	F-G	-0.09553	[-0.18198, -0.01658]	0.02148	0.32227
sparse clean	F-Q	-0.01835	[-0.15562, 0.00926]	0.34375	1.00000
sparse clean	F-G-L+N	-0.00044	[-0.00503, 0.00391]	0.75391	1.00000
block missing	F-L	-0.00496	[-0.00795, 0.00149]	0.34375	1.00000
block missing	F-G	-0.06999	[-0.09386, -0.03466]	0.00195	0.03125
block missing	F-Q	0.00128	[-0.00782, 0.01082]	0.75391	1.00000
block missing	F-G-L+N	-0.00186	[-0.00506, 0.00248]	0.34375	1.00000

All 16 planned contrasts are retained. The corresponding CSV contains full numerical precision, sign counts, sample size and mean differences.

5 Difficult-observation ablations: maximum error

Case	Contrast	Median difference	Marginal 95% CI	Sign p	Holm p
extreme	F-L	-0.00454	[-0.01929, 0.00157]	0.34375	1.00000
extreme	F-G	0.00263	[-0.10959, 0.17301]	0.75391	1.00000
extreme	F-Q	-0.06696	[-0.12910, -0.00587]	0.00195	0.03125
extreme	F-G-L+N	-0.00083	[-0.01066, 0.00812]	1.00000	1.00000
sparse noisy	F-L	0.00978	[0.00026, 0.02206]	0.10938	1.00000
sparse noisy	F-G	0.16010	[-0.02661, 0.30880]	0.10938	1.00000
sparse noisy	F-Q	-0.07692	[-0.11896, 0.06072]	0.34375	1.00000
sparse noisy	F-G-L+N	0.00251	[-0.00598, 0.01383]	0.34375	1.00000
sparse clean	F-L	0.00568	[-0.00648, 0.03043]	0.75391	1.00000
sparse clean	F-G	0.15141	[-0.25719, 0.36344]	0.75391	1.00000
sparse clean	F-Q	-0.08965	[-0.55148, -0.02004]	0.02148	0.32227
sparse clean	F-G-L+N	0.00115	[-0.01364, 0.02103]	0.75391	1.00000
block missing	F-L	0.00468	[-0.00191, 0.01363]	0.10938	1.00000
block missing	F-G	0.05695	[-0.06434, 0.11975]	0.75391	1.00000
block missing	F-Q	-0.06488	[-0.17483, 0.05398]	0.75391	1.00000
block missing	F-G-L+N	0.00533	[-0.00413, 0.01167]	0.10938	1.00000

All 16 planned contrasts are retained. The corresponding CSV contains full numerical precision, sign counts, sample size and mean differences.

6 Doubled-step ablations: L2

Case	Contrast	Median difference	Marginal 95% CI	Sign p	Holm p
helmholtz	F-L	-0.02571	[-0.04356, 0.00804]	0.10938	1.00000
helmholtz	F-G	-0.01177	[-0.03579, 0.02646]	0.34375	1.00000
helmholtz	F-Q	-0.01091	[-0.01696, 0.00680]	0.10938	1.00000
helmholtz	F-G-L+N	0.03543	[0.01198, 0.07751]	0.10938	1.00000
allen cahn	F-L	0.00105	[-0.00404, 0.00642]	0.75391	1.00000
allen cahn	F-G	0.00072	[-0.00409, 0.00376]	0.75391	1.00000
allen cahn	F-Q	0.00585	[-0.01123, 0.01091]	0.34375	1.00000
allen cahn	F-G-L+N	0.00322	[-0.00235, 0.01041]	0.34375	1.00000
burgers	F-L	0.01041	[-0.02607, 0.10724]	0.75391	1.00000
burgers	F-G	-0.03001	[-0.14295, 0.17147]	1.00000	1.00000
burgers	F-Q	0.01883	[-0.04312, 0.09074]	0.10938	1.00000
burgers	F-G-L+N	0.07100	[-0.03707, 0.34977]	0.75391	1.00000
wave	F-L	0.00158	[0.00057, 0.00336]	0.10938	1.00000
wave	F-G	-0.10001	[-0.14629, -0.08958]	0.00195	0.03125
wave	F-Q	0.00109	[-0.00036, 0.00280]	0.10938	1.00000
wave	F-G-L+N	-0.01386	[-0.02277, 0.00113]	0.10938	1.00000

All 16 planned contrasts are retained. The corresponding CSV contains full numerical precision, sign counts, sample size and mean differences.

7 Doubled-step ablations: maximum error

Case	Contrast	Median difference	Marginal 95% CI	Sign p	Holm p
helmholtz	F-L	-0.03847	[-0.05139, -0.02822]	0.02148	0.30078
helmholtz	F-G	-0.01649	[-0.05188, 0.00328]	0.10938	1.00000
helmholtz	F-Q	-0.00826	[-0.01515, -0.00038]	0.10938	1.00000
helmholtz	F-G-L+N	0.11349	[0.08393, 0.13949]	0.00195	0.03125
allen cahn	F-L	0.00047	[-0.00457, 0.00833]	0.75391	1.00000
allen cahn	F-G	-0.01222	[-0.02147, 0.00237]	0.10938	1.00000
allen cahn	F-Q	0.00777	[-0.00100, 0.01426]	0.10938	1.00000
allen cahn	F-G-L+N	0.00800	[0.00025, 0.01194]	0.10938	1.00000
burgers	F-L	0.01202	[-0.02989, 0.23255]	0.75391	1.00000
burgers	F-G	0.11224	[-0.07725, 0.35343]	0.75391	1.00000
burgers	F-Q	0.06338	[-0.00675, 0.24252]	0.34375	1.00000
burgers	F-G-L+N	0.01725	[-0.15208, 0.32778]	0.75391	1.00000
wave	F-L	0.00265	[-0.00032, 0.00652]	0.34375	1.00000
wave	F-G	-0.19593	[-0.30415, -0.14881]	0.00195	0.03125
wave	F-Q	0.00005	[-0.01243, 0.00539]	1.00000	1.00000
wave	F-G-L+N	-0.03552	[-0.04417, -0.01935]	0.02148	0.30078

All 16 planned contrasts are retained. The corresponding CSV contains full numerical precision, sign counts, sample size and mean differences.

8 Common-loss baselines: L2

Case	Contrast	Median difference	Marginal 95% CI	Sign p	Holm p
block missing	V-M5	-0.19580	[-0.24059, -0.18515]	0.00195	0.03125
block missing	V-M6	-0.19318	[-0.24042, -0.18069]	0.00195	0.03125
block missing	V-M7	-0.17172	[-0.20840, -0.12736]	0.00195	0.03125
block missing	V-VW	-0.13158	[-0.15465, -0.09959]	0.00195	0.03125
sparse clean	V-M5	-0.22439	[-0.32713, -0.15273]	0.00195	0.03125
sparse clean	V-M6	-0.21344	[-0.32406, -0.15757]	0.00195	0.03125
sparse clean	V-M7	-0.17495	[-0.24497, -0.13279]	0.00195	0.03125
sparse clean	V-VW	-0.13281	[-0.22278, -0.08119]	0.02148	0.10742
sparse noisy	V-M5	-0.19789	[-0.30269, -0.17423]	0.00195	0.03125
sparse noisy	V-M6	-0.18849	[-0.26506, -0.17001]	0.02148	0.10742
sparse noisy	V-M7	-0.15852	[-0.21516, -0.13597]	0.02148	0.10742
sparse noisy	V-VW	-0.14824	[-0.22659, -0.04931]	0.10938	0.10938
extreme	V-M5	-0.13299	[-0.20818, -0.06938]	0.00195	0.03125
extreme	V-M6	-0.13806	[-0.20866, -0.11213]	0.00195	0.03125
extreme	V-M7	-0.13349	[-0.15551, -0.06411]	0.00195	0.03125
extreme	V-VW	-0.10754	[-0.13560, -0.08287]	0.02148	0.10742

All 16 planned contrasts are retained. The corresponding CSV contains full numerical precision, sign counts, sample size and mean differences.

9 Common-loss baselines: maximum error

Case	Contrast	Median difference	Marginal 95% CI	Sign p	Holm p
block missing	V-M5	0.01235	[-0.06933, 0.09525]	0.75391	1.00000
block missing	V-M6	0.02269	[-0.06887, 0.10446]	0.75391	1.00000
block missing	V-M7	0.04613	[-0.05002, 0.11892]	0.34375	1.00000
block missing	V-VW	-0.00873	[-0.04619, 0.09590]	0.75391	1.00000
sparse clean	V-M5	0.11144	[-0.18579, 0.44740]	0.75391	1.00000
sparse clean	V-M6	0.15185	[-0.17668, 0.51708]	0.75391	1.00000
sparse clean	V-M7	0.09863	[-0.17485, 0.46107]	1.00000	1.00000
sparse clean	V-VW	0.11941	[-0.09695, 0.54083]	0.34375	1.00000
sparse noisy	V-M5	0.19446	[-0.11534, 0.57683]	0.75391	1.00000
sparse noisy	V-M6	0.23125	[-0.08381, 0.58432]	0.10938	1.00000
sparse noisy	V-M7	0.20019	[-0.06998, 0.45182]	0.75391	1.00000
sparse noisy	V-VW	0.21063	[-0.10367, 0.56513]	0.75391	1.00000
extreme	V-M5	0.00121	[-0.06042, 0.35473]	1.00000	1.00000
extreme	V-M6	-0.01976	[-0.05792, 0.33513]	0.75391	1.00000
extreme	V-M7	0.00441	[-0.03341, 0.29312]	0.75391	1.00000
extreme	V-VW	-0.04333	[-0.13082, 0.22889]	0.75391	1.00000

All 16 planned contrasts are retained. The corresponding CSV contains full numerical precision, sign counts, sample size and mean differences.

10 Independent-host reproduction protocol

The protocol reconstructs scores of all 560 archived V10 models and all 96 paired contrasts, then retrains 12 cases chosen before execution: seed 11, global-only and full-exponential for all four doubled-budget PDEs; noisy Burgers M5, M6, M7 and full-exponential at the V10 robustness budget. Selection uses the lowest declared seed, not favorable outcomes.

Frozen-score/table tolerance: relative 1e-8, absolute 1e-10. Retraining tolerance: relative 1e-3, absolute 1e-5 for L2 and maximum error, with exact matching initial tensor fingerprints and completed step counts. Discrepancies are retained and thresholds are not relaxed after observation.

The original runtime reports Python 3.12.13, PyTorch 2.14.0+cpu and NumPy 2.3.5. The portable public environment pins Python 3.12, PyTorch 2.8.0+cpu and NumPy 2.3.5, with SciPy 1.16.1 and Matplotlib 3.10.5. It therefore tests portability across both host and PyTorch version, not bitwise replication of an identical software stack.

Execution evidence is included in `submission/v11/reproduction/`.

The independent Ubuntu execution reproduced all 560 frozen scores and all 96 contrast statistics. The 55 automated tests passed. All 12 retrainings met the original numerical tolerances and completed the prescribed steps.

The complete predeclared gate passed for 8/12 cases; overall gate = False. Initial model fingerprints match in all 12 cases. Training/reference-set fingerprints differ in the four noisy-observation cases, so their strict gate remains failed.

The maximum absolute differences in final errors are 9.5275e-9 for L2 and 1.3997e-8 for maximum error. The original raw initial tensor arrays were not archived; hashes alone cannot identify the differing entries or establish the cause. A cross-version numerical effect is plausible but unproven. The original validation file remains unchanged.

Independent workflow and preserved failure. The complete 12-case comparison and original/new fingerprints are archived with this submission package.

11 Observation model details

For clean sensor values y, the scale is the maximum of their population standard deviation, mean absolute value and 1e-12. Gaussian corruption adds noise level times scale times independent standard-normal draws. Correlated corruption first applies z[j] = 0.8 z[j-1] + 0.6 e[j], initialized with z[0] = e[0]. This correlation follows sensor-array order; it is not a prescribed physical time-covariance model. Outliers add random signed shifts of 5 times max(noise level, 0.01) times scale to the configured fraction of entries. The missing-block case removes the middle 50 percent of 128 sensors ranked by time, retaining 64; the removed interval depends on the seed rather than a fixed pair of endpoints.

12 Limitations of the independent check

These 12 retrainings are numerical portability checks, not 12 new independent physical cases or a new confirmatory statistical campaign. They do not establish portability across all GPUs, compilers and accelerators. All scientific conclusions remain tied to the original controlled experiments and stated limitations.